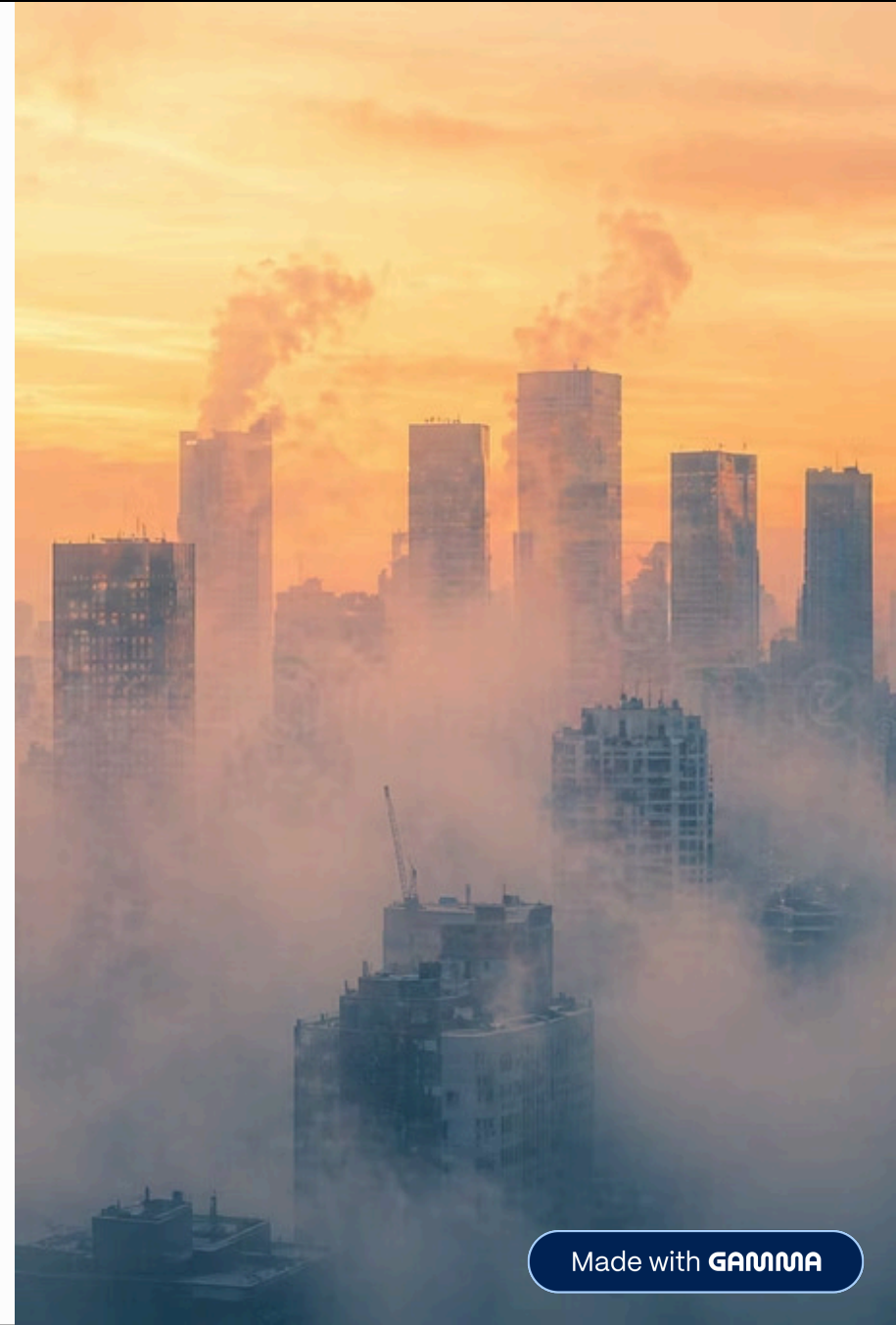


Beijing PM2.5 Analysis & Alert Optimization

Public Health & Environmental Monitoring



Project Team



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Context and Problem Statement ?

Sector Context

Air pollution is a severe public health hazard in rapidly developing urban centers, leading to significant respiratory illnesses and economic impact.

Stakeholder

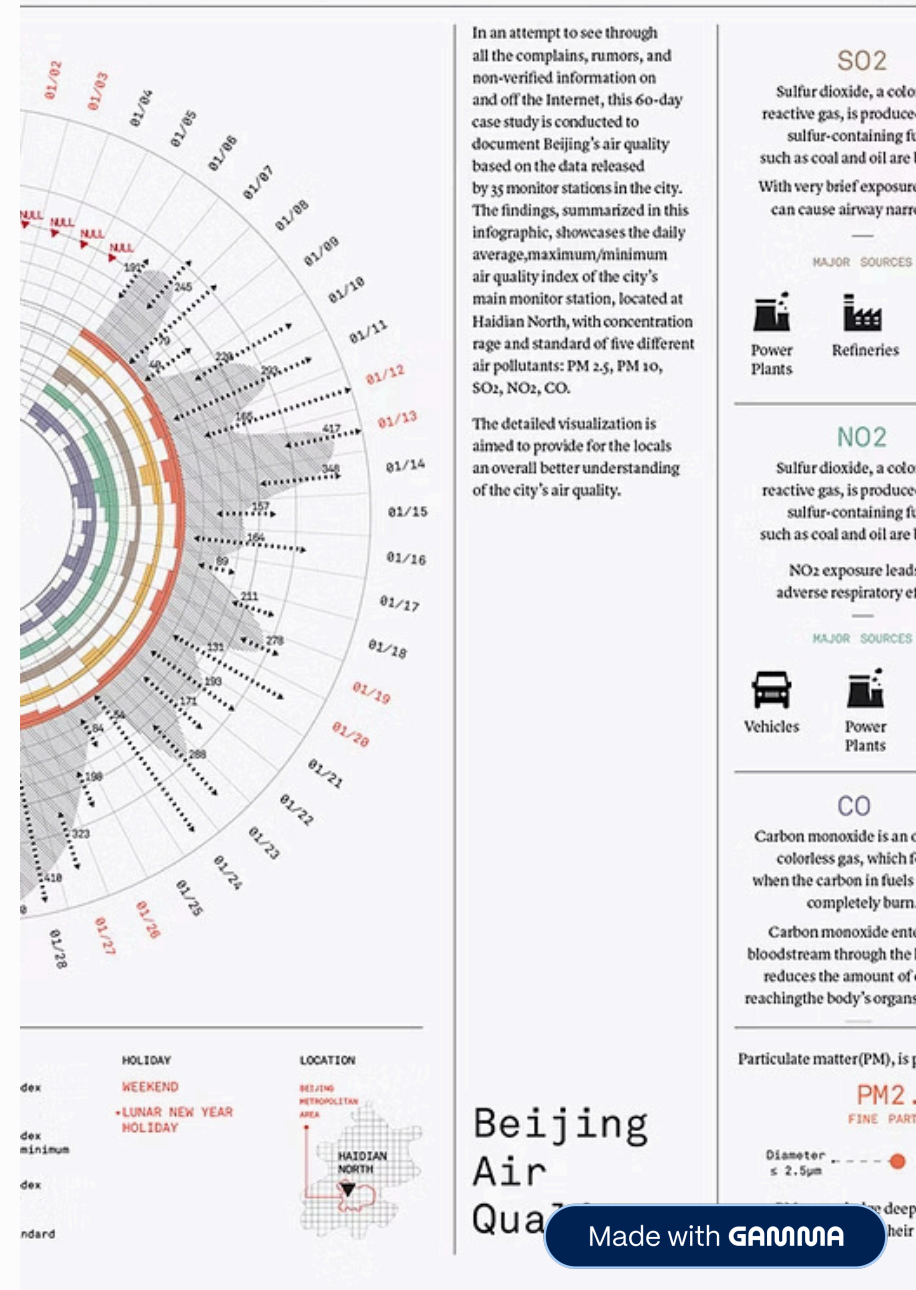
City Environmental Protection Bureau / Public Health Officials

Core Problem Statement

At what hours, months, and weather conditions does PM2.5 exceed unsafe levels most consistently, so authorities can decide when to trigger targeted pollution alerts and mitigation actions?

Decision Supported

This analysis will enable stakeholders to decide when to issue high-risk air-quality alerts and deploy targeted interventions based on the time periods and weather conditions most associated with unsafe PM2.5 levels.



Data Engineering ⚙️

01

Source

UCI Machine Learning Repository (US Embassy, Beijing)

03

Major Cleaning Steps

- Standardized datetime format for time-series compatibility
- Addressed missing PM2.5 values (~4.7%) by applying forward-fill (and backward-fill for leading nulls) to preserve timeline integrity without aggressive data dropping

02

Size and Coverage

43,824 rows across 5 years (Jan 1, 2010 – Dec 31, 2014) representing hourly granularity

04

Data Dictionary Summary

Key variables include pm2_5_cleaned (target variable), temperature, dew_point, pressure, wind_direction, and wind_speed

KPI Framework 🤖

Target KPI 1: Percentage of Hours with PM2.5 Above 150 $\mu\text{g}/\text{m}^3$

Definition: The frequency of extreme health risks in the recorded data.

Why it matters: Measures how often conditions become hazardous, guiding the necessity and volume of emergency alerts.

Target KPI 2: Average PM2.5 Concentration ($\mu\text{g}/\text{m}^3$)

Definition: The absolute baseline level of PM2.5 in the air.

Why it matters: Acts as the primary metric for tracking overall air quality trends against the WHO safe limit of 15 $\mu\text{g}/\text{m}^3$.

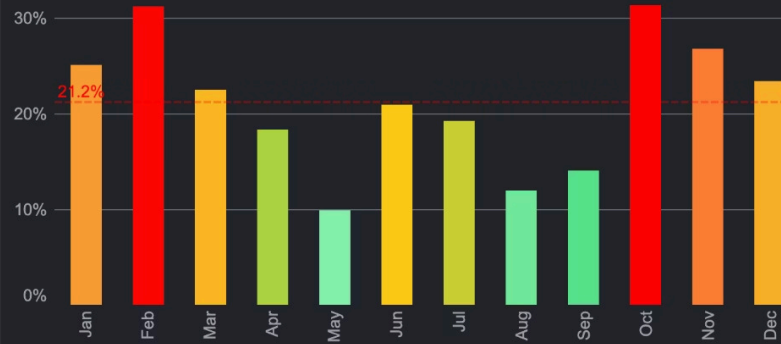
Mean PM2.5

97.8 $\mu\text{g}/\text{m}^3$
6.5x WHO limit

% Hours Above 150

21.2 %
~1 in 5 hours hazardous

% Hours Above 150 by Month



KPI Framework 🚫

Target KPI 3: Worst Month

Definition: The calendar month exhibiting the highest frequency of hazardous PM2.5 levels (e.g., Oct with 31.3% of hours).

Why it matters: Essential for strategic planning, allowing authorities to pre-allocate resources and schedule interventions before peak pollution seasons.

Target KPI 4: Peak Hour

Definition: The specific hour of the day recording the highest average PM2.5 concentration (e.g., 1 AM averaging 112.4 $\mu\text{g}/\text{m}^3$).

Why it matters: Crucial for daily operational planning, such as adjusting outdoor work schedules and broadcasting targeted health advisories to vulnerable populations.

📌 **Expected Insight:** We expect to find that PM2.5 levels are significantly higher during winter months and low-wind conditions, and lower during periods with stronger wind or rainfall, indicating a clear relationship between weather patterns and pollution intensity.

Worst Month

Oct

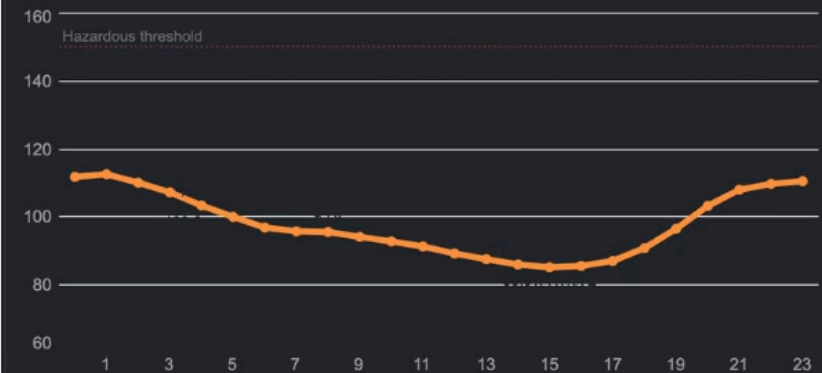
31.3% of hours

Peak Hour

1 AM

avg 112.4 $\mu\text{g}/\text{m}^3$

Hourly Average PM2.5



Key EDA Insights

1

Consistently High Baseline

The mean PM2.5 is $97.8 \mu\text{g}/\text{m}^3$ (over 6x the WHO limit), showing hazardous conditions are a recurring pattern.

2

Night-Time Spikes

Pollution peaks between midnight and 2 AM ($112.4 \mu\text{g}/\text{m}^3$) and reaches its lowest around 3–4 PM due to atmospheric mixing and temperature inversions.

3

Wind Direction Matters

Northwesterly winds clear pollution (avg $69.8 \mu\text{g}/\text{m}^3$), while calm/variable conditions allow it to accumulate (avg $124.5 \mu\text{g}/\text{m}^3$).

4

Winter Peaks

February and October have the highest PM2.5, strongly associated with increased coal heating and lower atmospheric dispersion.

5

Alarming AQI Category

Over half of all recorded hours (58.7%) fall into the "Unhealthy", "Very Unhealthy", or "Hazardous" categories.

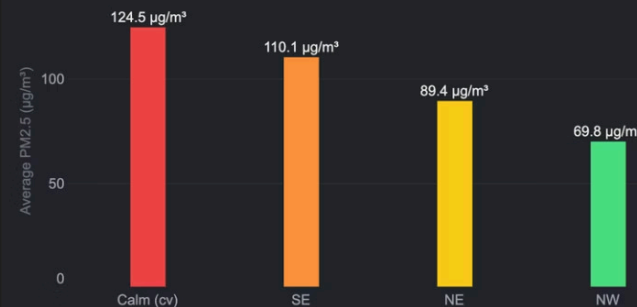
Mean PM2.5

97.8 $\mu\text{g}/\text{m}^3$
6.5x WHO limit

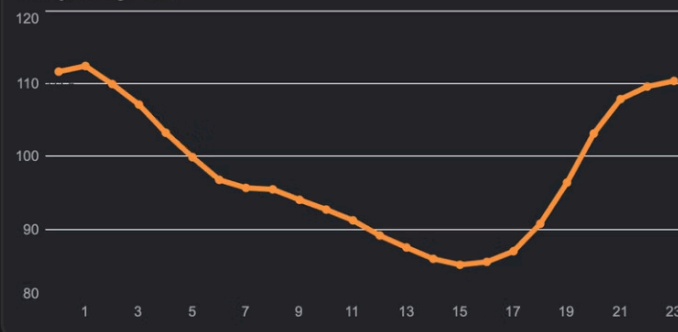
Peak Hour

1 AM
avg $112.4 \mu\text{g}/\text{m}^3$

PM2.5 by Wind Direction



Hourly Average PM2.5



% Hours Above 150 by Month



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Advanced Analysis 🧐

Statistical Method

Non-parametric tests were used (Shapiro-Wilk test confirmed PM2.5 is non-normally distributed). Applied Mann-Whitney U, Kruskal-Wallis, and Spearman correlations.

-0.35

Wind Speed Effect

Spearman correlation ($p < 0.001$)

Wind Speed Effect (Spearman Correlation)

Confirmed as the strongest negative predictor of PM2.5 ($r = -0.35$, $p < 0.001$); higher winds are consistently associated with lower pollution.

<0.001

Seasonal Significance

Kruskal-Wallis test

Seasonal Significance (Kruskal-Wallis)

Statistically significant differences across seasons ($p < 0.001$). Winter is confirmed as the highest pollution season, making it a reliable alert variable.

0.63

Rain Effect

Mann-Whitney U test

Rain & Snow Signals (Mann-Whitney U)

Rain has no statistically significant independent effect on PM2.5 ($p = 0.63$). Snow is correlated with higher PM2.5, but acts as a co-occurring winter signal rather than a direct cause.

Tableau Dashboard Walkthrough

<https://public.tableau.com/app/profile/milind.bansal5979/viz/DVA2-Capstone/RiskSeverityOverview?publish=yes>

PAGE 1 — RISK & SEVERITY OVERVIEW

How bad is it? AQI classification & hazardous day tracking

KPIs: Overall avg PM2.5 (97.8 $\mu\text{g}/\text{m}^3$) · Avg hazardous days / year (71.8 days) · Very Unhealthy + Hazardous (21.2%) · Peak PM2.5 recorded (994 $\mu\text{g}/\text{m}^3$)

Charts: AQI category breakdown · Hazardous days per year · Annual Trend

Filters: AQI Category · Year



PAGE 2 — TEMPORAL PATTERNS

When does pollution peak?

KPIs: Mean PM2.5 (97.8 $\mu\text{g}/\text{m}^3$) · % Hours Above 150 (21.2%) · Worst Month (Oct - 31.3% of hours) · Peak Hour (1 AM - avg 112.4 $\mu\text{g}/\text{m}^3$)

Charts: % Hours Above 150 by Month · Hourly Average PM2.5 · Hour x Month Heatmap

Filters: Hour · Year · Month



PAGE 3 — WEATHER & CONDITIONS

What drives pollution? Wind, season & precipitation

KPIs: Calm vs NW wind gap (54.7 $\mu\text{g}/\text{m}^3$) · Wind Speed Spearman r (-0.35) · Winter vs Spring gap (21.7 $\mu\text{g}/\text{m}^3$) · Rain Effect p-value (0.63)

Charts: PM2.5 by Wind Speed Range · PM2.5 by Wind Direction · PM2.5 by Season · Precipitation

Filters: Precipitation Type · Season · Wind Direction · Wind Speed



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Top Insights



Wind Speed is the Best Cleanser

Strong winds, particularly from the Northwest, disperse pollutants, whereas calm conditions allow rapid hazardous accumulation.



Precipitation is a False Trigger

Rainfall does not significantly clear PM2.5 pollution; tying alerts to rain forecasts is ineffective.



Pollution is Highly Seasonal & Diurnal

It spikes predictably in winter months and late-night hours due to coal heating and reduced atmospheric turbulence.



No Significant Improvement

Across the 5-year study window, annual means fluctuated but showed no consistent downward trend toward WHO guidelines.

Recommendations



Recommendation 1: Wind-Based Warning Systems

Implement automated public health warnings when sustained calm/variable winds are forecasted during winter, as this precedes hazardous accumulation.



Recommendation 2: Targeted Winter Interventions

Allocate peak resources (e.g., mask distribution, vulnerable population warnings) explicitly between October and February.



Recommendation 3: Restructure Night Shifts

For outdoor workers, avoid scheduling heavy outdoor activity between midnight and 3 AM when pollution hits its diurnal peak.



Impact

✓ Expected Outcome

- Reduced emergency room visits due to respiratory issues by enabling citizens to take pre-emptive protective measures before pollution spikes.
- Better allocation of municipal resources based on highly predictable seasonal/diurnal timing.

i Priority and Feasibility

- High Priority, High Feasibility. Meteorological data (wind speed/direction and season) is already tracked continuously, making an alert API easy to integrate.



Limitations 🤔

Single Monitoring Station

Data represents only central Beijing (US Embassy sensor) and cannot be accurately generalized to suburban or rural areas.

Confounding Variables

The dataset lacks economic or emission-source data (e.g., industrial output, traffic volume) which limits the ability to isolate exact causes of pollution spikes.

Cumulative Meteorological Readings

Rain and snow are recorded as cumulative hours, not instantaneous intensity, making complex weather modeling challenging.

Next Steps

To enhance the current analysis and provide a more comprehensive solution, we propose the following future extensions:



Multi-Station Spatial Analysis

Incorporate data from multiple monitoring stations across Beijing to create a dynamic spatial heatmap of pollution, providing a granular view of localized PM2.5 concentrations and their spread across different districts.



Emission Source Correlation

Integrate proxy data for key emission sources, such as traffic density, industrial outputs, and residential heating fuel consumption, to conduct a more robust root-cause analysis and directly link pollution spikes to specific activities.

Closing Summary

While Beijing's PM2.5 challenge remains significant, our analysis reveals strong correlations between pollution levels and predictable meteorological patterns (wind, season, diurnal cycles). These insights pave the way for a highly predictive, data-driven early warning system, enabling proactive public health interventions and resource allocation.